

Mud Pump Specifications & Operating Reference

National 12-P-160 / Gardner Denver PZ-11 / National 10-P-130

Permian Basin Drilling Operations

1. Pump Specifications Summary

1.1 National 12-P-160 (Primary Pump — Rig Nos. 100–149)

Parameter	Value
Type	Triplex single-acting piston
Maximum input power	1,600 HP
Maximum stroke rate	120 SPM
Maximum working pressure	7,500 psi
Liner sizes available	4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0 in
Stroke length	12 in
Rod load rating	162,000 lbf
Fluid end material	4140 steel, chrome-lined bores
Suction connection	8 in flanged
Discharge connection	4 in flanged

Output volume by liner size (gal/stroke):

Liner (in)	Vol (gal/stroke)	Max flow @ 120 SPM (gpm)
4.0	0.980	117.6
4.5	1.241	148.9
5.0	1.531	183.7
5.5	1.852	222.2
6.0	2.203	264.4
6.5	2.584	310.1
7.0	2.994	359.3

Maximum pressure by liner size:

Liner (in)	Max pressure (psi)
4.0	7,500
4.5	7,500
5.0	6,000
5.5	5,000
6.0	4,200
6.5	3,500
7.0	3,000

1.2 Gardner Denver PZ-11 (Secondary Pump — Rig Nos. 150–199)

Parameter	Value
Type	Triplex single-acting piston
Maximum input power	1,100 HP
Maximum stroke rate	110 SPM
Maximum working pressure	7,500 psi
Liner sizes available	4.0, 4.5, 5.0, 5.5, 6.0, 6.5 in
Stroke length	11 in
Rod load rating	148,000 lbf

Output volume by liner size (gal/stroke):

Liner (in)	Vol (gal/stroke)	Max flow @ 110 SPM (gpm)
4.0	0.899	98.9
4.5	1.138	125.2
5.0	1.405	154.6
5.5	1.700	187.0
6.0	2.022	222.4
6.5	2.372	260.9

2. Liner & Piston Selection Guide

2.1 Selection Criteria

- Select liner size to achieve required flow rate at $\leq 80\%$ of maximum SPM
- Verify selected liner pressure rating exceeds maximum anticipated standpipe pressure

- For Wolfcamp A laterals: typical requirement is 400–600 gpm at 3,000–4,500 psi

2.2 Recommended Liner for Permian Basin Lateral Drilling

- **5.5 in liner:** 400–500 gpm range, pressure rating 5,000 psi — preferred for 6-1/8 in hole
- **6.0 in liner:** 500–600 gpm range, pressure rating 4,200 psi — preferred for 8-3/4 in hole
- **5.0 in liner:** high-pressure applications >4,500 psi standpipe

2.3 Piston Types

Piston Type	Application	Service Life
Rubber bonded	WBM, low-abrasion	200–400 hours
Urethane	OBM, high-abrasion	400–800 hours
Bi-directional	High-pressure >5,000 psi	150–300 hours

3. Pump Efficiency & Volumetric Output

3.1 Pump Efficiency

- New pump with new liners/pistons: 98–99% volumetric efficiency
- Worn liners (>300 hours): 92–96%
- Worn pistons: 88–94%
- Rule of thumb: assume 95% efficiency for flow calculations

3.2 Actual Flow Rate Calculation

$$Q_{\text{actual}} \text{ (gpm)} = \text{SPM} \times \text{Vol_per_stroke (gal)} \times \text{Efficiency}$$

3.3 Lag Time Calculation (Bottoms-Up)

$$\text{Lag_strokes} = \text{Annular_volume (bbl)} / \text{Output_per_stroke (bbl/stroke)}$$

$$\text{Lag_time (min)} = \text{Lag_strokes} / \text{SPM}$$

- 1 bbl = 42 gallons

4. Maintenance Procedures

4.1 Daily Checks



- ☐ Liner wash fluid level and condition (should be clean, no mud contamination)
- ☐ Piston rod packing — check for leaks at stuffing box
- ☐ Suction and discharge valve covers — no leaks
- ☐ Dampener pre-charge pressure (nitrogen): should be 1/3 of operating pressure
- ☐ Gear oil level in power end
- ☐ Crosshead pin lubrication

4.2 Liner & Piston Inspection Intervals

- **Every 100 hours:** Inspect piston condition; measure liner bore for wear
- **Every 200 hours:** Pull and inspect suction/discharge valves and seats
- **Every 500 hours:** Full fluid end inspection; replace all expendables

4.3 Liner Wear Limits

- Maximum liner bore wear: +0.010 in from nominal
- If bore is out-of-round by >0.005 in: replace liner
- Scoring or pitting >1/16 in deep: replace liner

4.4 Valve & Seat Replacement

Symptoms requiring immediate valve inspection: - Pump pressure fluctuation >200 psi at constant SPM - Knocking or chattering noise from fluid end - Reduced output at same SPM (>5% drop in efficiency)

Valve replacement procedure: 1. Shut down pump; bleed pressure from discharge line 2. Remove valve cover (torque: 250–300 ft-lbs) 3. Extract valve assembly using valve puller tool 4. Inspect seat for erosion, pitting, or cracking 5. Replace valve and seat as a matched set 6. Torque valve cover to spec; pressure test to 500 psi before returning to service

4.5 Piston Rod Packing Adjustment

- Acceptable leak rate: 1–3 drops per minute at operating pressure
- If leaking >10 drops/min: tighten packing gland 1/4 turn; recheck
- If leaking persists after 3 adjustments: replace packing

5. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Pressure fluctuation ± 200 psi	Worn/damaged valve	Inspect and replace valves

Symptom	Probable Cause	Corrective Action
Low output at normal SPM	Worn liner or piston	Measure liner; replace piston
Knocking from fluid end	Cavitation or valve failure	Check suction pressure; inspect valves
Overheating power end	Low gear oil or bearing failure	Check oil level; inspect bearings
Piston rod leak	Worn packing	Adjust or replace packing
Suction pressure <15 psi	Plugged suction screen	Clean suction screen
Pump won't reach rated pressure	Worn liners, piston bypass	Replace fluid end expendables

6. Hydraulics Reference

6.1 Equivalent Circulating Density (ECD)

$$\text{ECD (ppg)} = \text{MW} + (\text{Annular_Pressure_Loss_psi} / (0.052 \times \text{TVD_ft}))$$

6.2 Annular Velocity

$$\text{AV (ft/min)} = (24.51 \times \text{Q_gpm}) / (\text{D_hole}^2 - \text{D_pipe}^2)$$

Where D_hole and D_pipe are in inches.

Minimum AV for cuttings transport: - Vertical: 100–150 ft/min - Deviated (>45°): 150–200 ft/min - Horizontal: 200–300 ft/min (with rotation)

6.3 Bit Hydraulics — Nozzle Sizing

$$\text{HHP_bit} = (\text{P_standpipe} \times \text{Q_gpm}) / 1714$$

$$\text{Nozzle_area (in}^2\text{)} = \text{Q_gpm} / (\text{C_d} \times \sqrt{(\text{P_nozzle_psi} \times 12,032)})$$

- C_d (discharge coefficient) = 0.95 for standard nozzles
- Target: 3.5–5.0 HHP/in² for PDC bits in Wolfcamp A